FOODHUB - ONLINE FOOD ORDERING AND DELIVERY SYSTEM

INDUSTRY INTERNSHIP REPORT

IT23723-INTERNSHIP II

Submitted by

MADHAVAN P (241001131)

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY





DEPARTMENT OF INFORMATION TECHNOLOGY

RAJALAKSHMI ENGINEERING COLLEGE

JULY 2026

BONAFIDE CERTIFICATE

Certified that this project report titled **"FOODHUB - ONLINE FOOD ORDERING AND DELIVERY SYSTEM"** is the bonafide work of **MADHAVAN P (241001131)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE OF THE REVIEWER

SIGNATURE OF THE MENTOR **SIGNATURE OF THE CLASS IN-CHARGE**

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Dr.P.Valarmathie
Head of The Department
Department of Information Technology
Rajalakshmi Engineering College

Submitted to Project Viva Voce Examination held on _____

ACKNOWLEDGEMENT

First, we thank the almighty God for the successful completion of the project. Our sincere thanks to our chairman **Mr. S. Meganathan, B.E., F.I.E** for his sincere endeavour in educating us in his premier institution.

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MADHAVAN P

ABOUT THE INTERNSHIP ORGANIZATION - TVK TECHNOLOGIES

TVK Technologies Ltd. is an Information Technology (IT) company located in Velachery, Chennai, Tamil Nadu. The organization provides software development services, corporate training, internship programs, and placement support for students and fresh graduates. Its mission is to bridge the gap between academic learning and industry requirements by offering practical, industry-oriented training.

The company offers internships in various domains such as Full Stack Development, Front-End Development, AWS with DevOps, Artificial Intelligence, Machine Learning, Data Science, Power BI, Digital Marketing, and Human Resources. The internship program focuses on hands-on learning through real-time projects, enabling students to develop technical, analytical, and problem-solving skills.

TVK Technologies also provides placement assistance, including technical training, mock interviews, resume preparation, and career guidance to improve students' employability. With experienced mentors and an industry-focused approach, the organization helps interns gain practical knowledge and prepares them for successful careers in the IT industry.

Organization Details:

- **Organization Name:** TVK Technologies Ltd.
- **Industry:** Information Technology (IT)
- **Location:** Velachery, Chennai, Tamil Nadu
- **Services:** Software Development, Corporate Training, Internship, and Placement Support
- **Website:** www.tvktechs.com

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LIST OF ABBREVIATIONS

ABBREVIATION	ACRONYM
API	Application Programming Interface
JWT	JSON Web Token
REST	Representational State Transfer
JPA	Java Persistence API
DTO	Data Transfer Object
CORS	Cross-Origin Resource Sharing
MVC	Model View Controller
UI/UX	User Interface / User Experience
JSON	JavaScript Object Notation
H2	In-Memory Relational Database
DOM	Document Object Model

ABSTRACT

The **FoodHub - Online Food Ordering and Delivery System** is a web-based application developed to simplify and automate the process of discovering restaurants, ordering meals, and managing deliveries within an organization or commercial ecosystem. Traditional food ordering methods, which rely on manual paper records, direct phone calls, or fragmented communication, are often time-consuming, error-prone, and inefficient. This system provides a centralized platform for managing customers, restaurant owners, menus, carts, order dispatches, and delivery status tracking.

The application is developed using Java (Spring Boot) for the backend, React.js (Vite) and CSS for the frontend, and an in-memory H2 database with JPA/Hibernate object-relational mapping. It follows a RESTful architecture with secure JWT-based authentication and role-based access control for customers, restaurant partners, and system administrators. The system enables users to search for outlets, customize cart orders, apply promo coupons, track live order history, and manage merchant dashboards.

To enhance efficiency, the system incorporates smart features such as cross-restaurant cart conflict detection, interactive map pin drop delivery address selection, dynamic coupon engines, predictive checkout calculations, and real-time order status updates. These features improve food service delivery, reduce manual effort, increase operational transparency, and support informed management decisions.

The FoodHub - Online Food Ordering and Delivery System provides a scalable, secure, and user-friendly solution for digital food e-commerce, making it suitable for modern food delivery networks seeking efficient operational control.

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1. Introduction

The FoodHub - Online Food Ordering and Delivery System is a web-based application developed to simplify and automate the management of food ordering and delivery services. It provides a centralized platform for maintaining restaurant listings, menu items, customer carts, order dispatches, discount offers, and system delivery metrics. The system replaces traditional manual methods, reducing errors and improving efficiency.

Developed using Java (Spring Boot), React.js, HTML, CSS, JavaScript, and H2 Database, the application offers secure user authentication, role-based access control, cart conflict warnings, map-based location selector, dashboard metrics, and status tracking. By digitizing the food ordering process, the system enhances transparency, improves resource utilization, and enables food merchants to manage their operations more effectively.

2. Objectives

The main objectives of this project are:

1. To develop a web-based system for efficient ordering and tracking of food deliveries.
2. To maintain a centralized database of restaurants, menus, orders, and customer address information.
3. To automate cart management, cart conflict warnings, coupon verification, and order processing.
4. To provide secure user authentication and role-based access control.
5. To generate analytics and operational dashboards for merchants and administrators for effective monitoring.
6. To reduce manual errors, improve operational efficiency, and ensure better delivery speed for consumers.

3. Existing System

In many food establishments and traditional order environments, food requests are handled using manual methods such as paper receipts or phone call recordings. These approaches require significant manual effort and are prone to order taking errors, duplicate records, and misplaced items. Tracking order dispatches, delivery statuses, and daily sales details becomes difficult as the number of orders increases. Additionally, existing manual systems lack real-time status monitoring, centralized data management, secure access control, and automated checkout calculations, making the overall food delivery management process inefficient and time-consuming.

4. Proposed System

The FoodHub - Online Food Ordering and Delivery System is a web-based application that provides a centralized and automated solution for managing online food orders. It enables administrators and restaurant owners to maintain outlet records, add menu items, process incoming customer orders, track dispatches, and monitor sales metrics efficiently. The system incorporates secure user authentication, role-based access control, Swiggy/Zomato-style cart conflict prompts, interactive map pin drop location selection, and live order tracking notifications. By replacing manual processes with a digital platform, the proposed system improves accuracy, enhances productivity, reduces administrative effort, and ensures seamless delivery workflows.

5. Modules

Authentication Module

Provides secure user login, registration, JWT-based authentication, and role-based access control for Customers, Owners, and Admins.

Outlets & Menu Explorer Module

Displays 25+ registered food outlets with categories (Biryani, Pizza, Desserts, South Indian) and dynamic menu showcases.

Cart & Conflict Resolution Module

Manages cart item addition, quantity adjustments, item removal, and multi-restaurant cart replacement warnings.

Coupon & Checkout Module

Validates discount coupons (e.g. SWIGGYIT, ZOMATO50) and calculates subtotals, delivery fees, and taxes in real time.

Delivery & Address Management Module

Handles saved delivery addresses and map pin drop coordinate selections for precise customer delivery positioning.

Merchant & Admin Dashboard Module

Allows restaurant owners to track active orders and update item availability while giving admins overall platform control.

6. Technologies Used

Technology	Purpose
Java 21	Backend application runtime development
Spring Boot 3.3	REST API framework and backend dependency injection
Spring Security & JWT	User authentication and role authorization filtering
Hibernate & JPA	Database connectivity and object-relational mapping
H2 Database	In-memory relational database for storing application data
React.js (Vite)	Frontend client rendering framework
Vanilla CSS3	User interface styling and royal purple responsive layout
Axios	API endpoint call communication client
Git & GitHub	Version control and source code management

8. Advantages / Features

- Centralized management of food outlets, menus, and orders.
- Secure login with JWT role-based access control.
- Swiggy/Zomato royal purple UI theme for optimal user experience.
- Efficient cart management with multi-kitchen conflict interception.
- Real-time dashboard with order status monitoring and sales analytics.
- Reduces manual entry errors and improves order fulfillment speed.

9. Learning Outcomes

Through this project, I gained:

- Gained practical experience in developing a full-stack web application using Java, Spring Boot, React.js, and CSS.
- Acquired knowledge of RESTful API development, DTO patterns, and JWT-based security authorization.
- Improved understanding of database entity mapping, JPA lazy loading resolution, and CRUD operations.
- Enhanced problem-solving, debugging, and software architecture implementation skills during internship.
- Knowledge of Git and GitHub version control workflows.

10. Future Enhancements

- Develop a native mobile application for Android and iOS to enable food ordering on mobile devices.
- Integrate live GPS rider tracking using Google Maps API for real-time delivery tracking.
- Incorporate online payment gateways like Razorpay or Stripe for automated digital transactions.
- Add AI-based recommendation engines to suggest food dishes based on previous order history.

SDG MAPPING

IT23723-Internship II

Project Title: FOODHUB - ONLINE FOOD ORDERING AND DELIVERY SYSTEM

SDG Goal	Activity addressing SDG Theme
SDG 9- Industry, Innovation and Infrastructure	Promotes digital transformation by replacing manual ordering management with a secure, scalable, and technology-driven web application.
SDG 12- Responsible Consumption and Production	Improves food demand forecasting by tracking orders, reducing unnecessary inventory purchases, minimizing wastage, and optimizing delivery routes.
SDG 8- Decent Work and Economic Growth	Enhances business productivity by automating order management processes, reducing manual effort, and improving operational efficiency.
SDG 16- Peace, Justice and Strong Institutions	Ensures transparency and accountability through secure authentication, audit logs, and role-based access control for order management.

CONCLUSION

The FoodHub - Online Food Ordering and Delivery System provides an efficient, secure, and user-friendly solution for managing food orders digitally. It simplifies outlet exploration, cart additions, discount applicators, delivery address pinning, and merchant order management while reducing manual effort and improving data accuracy. The project demonstrates the effective use of Java, Spring Boot, React.js, CSS, and H2 database to develop a modern e-commerce application. Overall, the system enhances organizational productivity, ensures smooth delivery operations, and serves as a scalable foundation for future enhancements.

APPENDIX

1. TVK Technologies Offer Letter

- **Position:** Full Stack Java Intern
- **Date of Issue:** 10.06.2026
- **Internship Period:** 10 June 2026 to 10 July 2026

2. Attendance & Completion Certificate

- **Verification Date:** 16.07.2026
- **Program Manager:** Priyatharshan A
- **Outcome:** Successful completion of Industrial Training / Internship in Java Full Stack Development.

— EMPOWERING CAREER GROWTH —

TVK Technologies

Internship Completion Letter

Date: 13/07/2026

This is to certify that **Mr. MADHAVAN R (241001132),** a student of **Rajalakshmi Engineering College** has successfully completed the internship in **Java Full Stack Development** at **TVK Technologies** from **10 June 2026 to 10 July 2026** under the supervision of **PRIYATHARSHAN A / Program Manager**.

During the internship period, he worked on the project titled **"Office Asset Management System"** and demonstrated sincerity, dedication, and a strong willingness to learn. He actively contributed to project development, task execution, documentation, and team collaboration throughout the internship.

His performance was professional, punctual, and well-structured. He displayed good problem-solving abilities, adaptability, and teamwork skills, which were appreciated by the organization.

We wish him continued success in all future endeavors and look forward to supporting his career journey.

Best wishes,

CEO / FOUNDER

MSME
MICRO, SMALL & MEDIUM ENTERPRISES

PROGRAM MANAGER

Phone: +91 94439 38850 **Website**: www.tvktechs.com **Email**: hr@tvktechs.com